

Game Theory and Social Choice

Lecture 12: The Intuitive Criterion

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Canadian Signal

Where We Left Off

Perfect Bayesian equilibrium put four constraints on players, and the fourth one was really weak.

If you reach a probability zero node, form some consistent beliefs, any consistent beliefs.

That weakness let the college game keep an equilibrium where employers hire beach-goers and reject graduates.

Today we look at one proposal for tightening it.

The Bar

A North American tourist walks into a bar. Of the North Americans who pass through, 60% are Canadian and 40% are American.

She knows her own nationality. The barman does not, and her accent gives nothing away.

She can ask for the TV to be turned to hockey or to baseball.

She knows that once she asks, the barman will form a view about where she is from.

Her Preferences

If she is American she has a weak preference for baseball. If she is Canadian she has a weak preference for hockey.

Either way she has a strong preference for being taken for a Canadian, stronger than her preference about the sport.

The barman wants to guess right.

All of this is common knowledge.

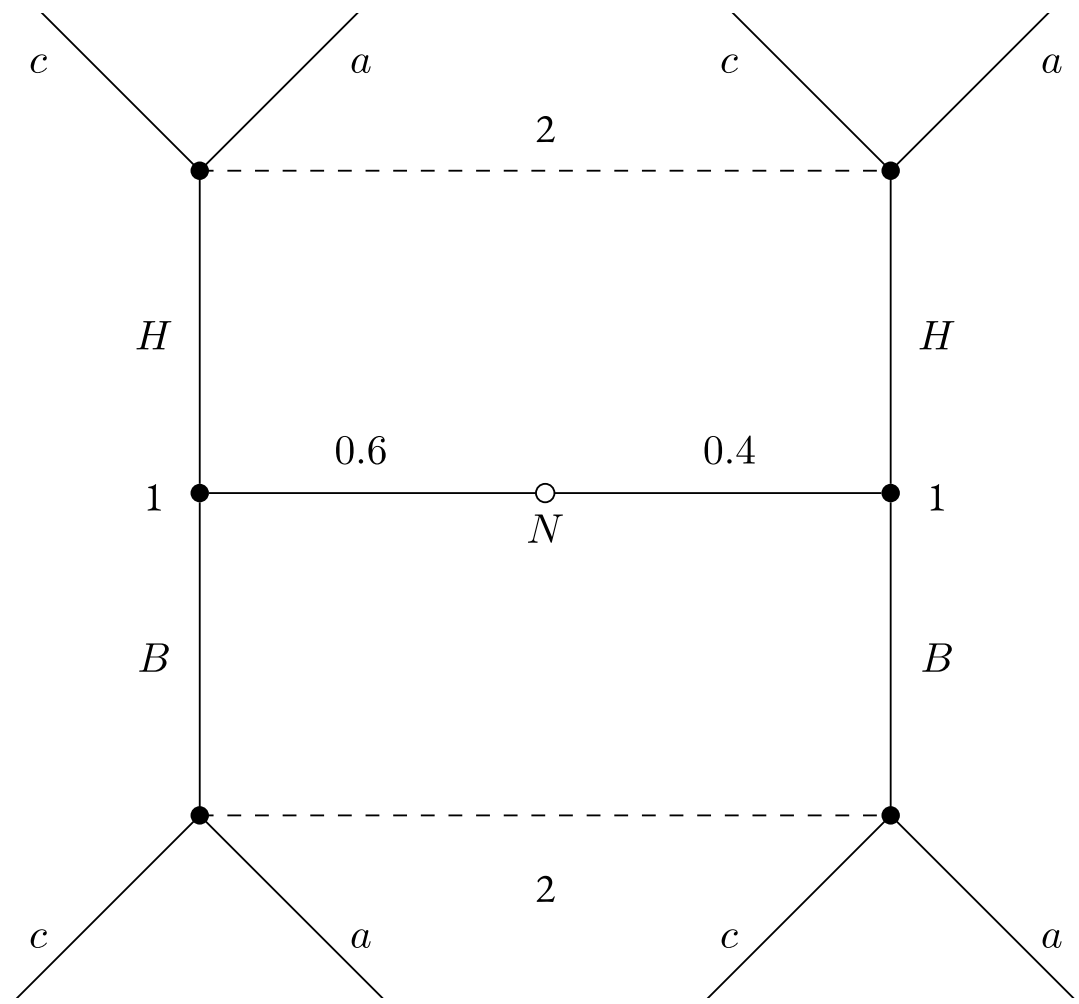
The Standard Version

Cho and Kreps tell this with beer and quiche, and a set of stereotypes about which sort of man orders which for breakfast. You can do the translation easily enough.

Their split is 90/10 rather than 60/40.

All that matters is that the people in the bar have a prior above 0.5 in the direction the stranger wants.

The Tree



The Tree

B is baseball, H is hockey. For the barman, a is guessing American and c is guessing Canadian.

Reading the Payoffs

The tourist gets 2 for being taken as Canadian, and 1 for the sport she prefers.

So her best outcome is 3 and her worst is 0.

A Canadian who asks for hockey and is taken for a Canadian gets 3.

An American who asks for baseball and is taken for a Canadian also gets 3.

The barman gets 1 for a correct guess and -1 for a wrong one.

Strategies

Player 1's strategy says what she does if Canadian and what she does if American. We write this as $\alpha\beta$, in that order.

Player 2's strategy says what he guesses after hockey and what he guesses after baseball. We write this as $\gamma\delta$, in that order.

So ca means: Canadian if hockey, American if baseball.

The Sensible Equilibrium

$\langle HH, ca \rangle$. Everyone asks for hockey, and the barman reads hockey as Canadian.

The Canadian gets 3, her maximum. The American gets 2, and switching to baseball would get her 1, so she stays.

The barman sees hockey from everyone, so he learns nothing and falls back on the prior.

Since 0.6 exceeds a half, he guesses Canadian.

The Odd Equilibrium

$\langle BB, ac \rangle$. Everyone asks for baseball, and the barman reads *baseball* as Canadian and hockey as American.

The American gets 3, her maximum. The Canadian gets 2; switching to hockey would have her taken for an American, which gets her 1.

And again the barman learns nothing from a request everyone makes, so guessing Canadian is right.

Why It Survives

For this to be a perfect Bayesian equilibrium, the barman has to be maximising at the hockey node too, which nobody ever reaches.

It works provided that, conditional on hockey, his credence that she is American climbs above a half.

Nothing we have said forbids that. So the equilibrium exists.

But it's really strange.

Happy, Content, Unhappy

Outcomes

Write an outcome as a pair $\langle P, N \rangle$: the players' moves together, and nature's move.

Fix N , and sort the players into three sorts.

The Three Sorts

Happy: given N , no other combination of players' moves would do better for them. They are at their ceiling.

Content: not happy, but no alternative strategy of their own would do better, given what nature and everyone else would do.

Unhappy: neither of the above.

What the Concepts Say

Nash equilibrium is the condition that nobody is unhappy.

So in any Nash equilibrium every player is happy or content, and the division between those two is doing no work yet.

The proposal is to put it to work on off-path beliefs.

The Weak Version

Suppose you are certain the combination P will be played. Now condition on some other player playing s' , which is not what they play in P .

If it is consistent with everything else you believe that s' was played by someone who is merely *content* with the actual outcome, give probability 1 to that.

In short: when somebody deviates, assume it was one of the content players rather than one of the happy ones.

Why That Is Plausible

Someone who is already at their ceiling has nothing to gain by wandering off.

Someone who is merely content has an incentive that is, at best, balanced.

A small change in circumstances flips them. (Maybe, like in Harsanyi's model, we weren't quite sure what their payoffs were.)

So if a deviation happens, the content player is the one who had a reason to make it.

Sorting the Bar

Take the odd equilibrium, $\langle BB, ac \rangle$.

The American asks for baseball and is taken for a Canadian, which is 3. Given that she is American, nothing does better. She is happy.

The Canadian gets 2. Her ceiling is 3, which she would reach by asking for hockey and being believed. So she is not happy. Switching to hockey gets her 1, so she is content.

What the Barman Must Think

A hockey request is a deviation.

By the criterion, the barman gives probability 1 to it coming from the content player.

The content player is the Canadian.

So he should respond to hockey by guessing Canadian, which is the opposite of what the odd equilibrium requires of him.

And Then It Unravels

If hockey gets read as Canadian, the Canadian tourist does better by asking for hockey: 3 rather than 2.

So she is no longer content with baseball, and $\langle BB, ac \rangle$ is no longer an equilibrium at all.

The sensible equilibrium survives, since there the deviation story runs the other way.

An Exercise

Last time's odd equilibrium was $\langle DD, rl \rangle$, with $p > \frac{1}{2}$: everybody at the beach, and the employer hiring beach-goers and rejecting graduates.

Does the criterion kill it?

It Does Not

In that equilibrium the high type gets 4, and so does the low type.

Four is the most either type can get anywhere in the game. Both of them are happy, and neither is merely content.

So there is no content player to pin a deviation on, and the criterion says nothing at all.
 $\langle DD, rl \rangle$ survives it.

What Does Kill It

A different and simpler constraint: don't read a deviation as coming from someone for whom it is strictly dominated.

College is strictly dominated for the low type. The worst the beach can do for her is 0, the best college can do is -1 .

It is not dominated for the high type, whose best from college is 3 against a worst case of 0. So a degree has to be read as high, and the employer has to hire.

A Weakness

$\langle DD, rl \rangle$ is exactly the sort of equilibrium a refinement ought to reject, and Cho & Kreps haven't rejected it.

The criterion works by comparing what you get in equilibrium with what you could get by deviating. Once $p > \frac{1}{2}$ the employer will hire an unsorted pool, so both types are already at their ceiling.

A criterion that compares payoffs has nothing left to compare.

This isn't an argument that they are *wrong*, but that their theory is *incomplete*.

Where It Does Work

Drop p below $\frac{1}{2}$ and take the pooling equilibrium $\langle DD, rr \rangle$: nobody goes, nobody is hired, everybody gets 0.

Now the types come apart. College could have got the high type 3, so she was only ever content to skip it. It could never have got the low type more than -1 , so she is happy to have skipped it.

A degree must therefore be read as high, the employer hires, and the high type gets 3 rather than 0. The pooling equilibrium breaks, and the separating one survives.

The Full Version

Two Options Is a Special Case

Where every player has two moves, the version above is equivalent to what Cho and Kreps actually wrote.

With three or more moves it is too blunt, because being happy is not a single verdict.

You can be happy to have avoided one alternative and only content to have avoided another.

Pairwise Instead

So redo the categories as comparisons between s and a particular alternative s' .

Happy to have played s rather than s' : the payoff from s beats every outcome s' could have produced, given N .

Content to have played s rather than s' : not happy in that sense, but s does at least as well as s' given N and the other players' strategies.

Equilibrium Dominance

When a player is happy to have played s rather than s' , and s belongs to an equilibrium outcome, Cho and Kreps call s' **equilibrium-dominated**.

The full criterion then says: conditional on seeing s' , give probability 1 to its coming from someone who was merely content to avoid it, rather than someone for whom it was equilibrium-dominated.

Where the Difference Shows

Suppose a player has moves s_1, s_2, s_3 , and is type t_1 or t_2 . In equilibrium both types play s_1 .

Type t_1 is happy to have played s_1 rather than s_2 , and merely content rather than s_3 . Type t_2 is the other way round.

On the weak version both types are merely content overall, so nothing is ruled out. On the full version, seeing s_2 should convince you the player is t_2 , and seeing s_3 should convince you she is t_1 .

The Shape of a Refinement

What the criterion constrains is a conditional credence, conditional on an event the agent gives probability zero.

Every refinement in this area has that shape, though they disagree about what the constraint should be.

A Large Literature

Google Scholar has over five thousand citations for the Cho and Kreps paper, and this *kind* of response is common enough in the ones I've surveyed. (Concession: Not all 5000 of them.)

There are arguments that the criterion is too weak and arguments that it is too strong, and refinements running in both directions. But we'll stop here rather than digging into the refinement literature.

Back to Philosophy

Updating on a proposition you had assigned probability zero is a problem philosophers have worried about, usually in the abstract.

So there are all these papers recently on things like, a dart is thrown at a dartboard, and lands with precisely $\langle x, y \rangle$ coordinates; what is your updated credence on learning the true value for x .

It's all a bit abstract, and the results are a bit of a mess.

Here is a literature that has worried about it with specific cases, specific proposals, and an agreed standard for when a proposal fails.

On this particular question the economists have done more work than the philosophers.

Next Time

We're leaving the math behind, and asking when the math models anything in the real world.

In particular, we're going to look at Spence's model, and ask whether it is a plausible model of colleges as we actually find them.